



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI CHELLIAMMAN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

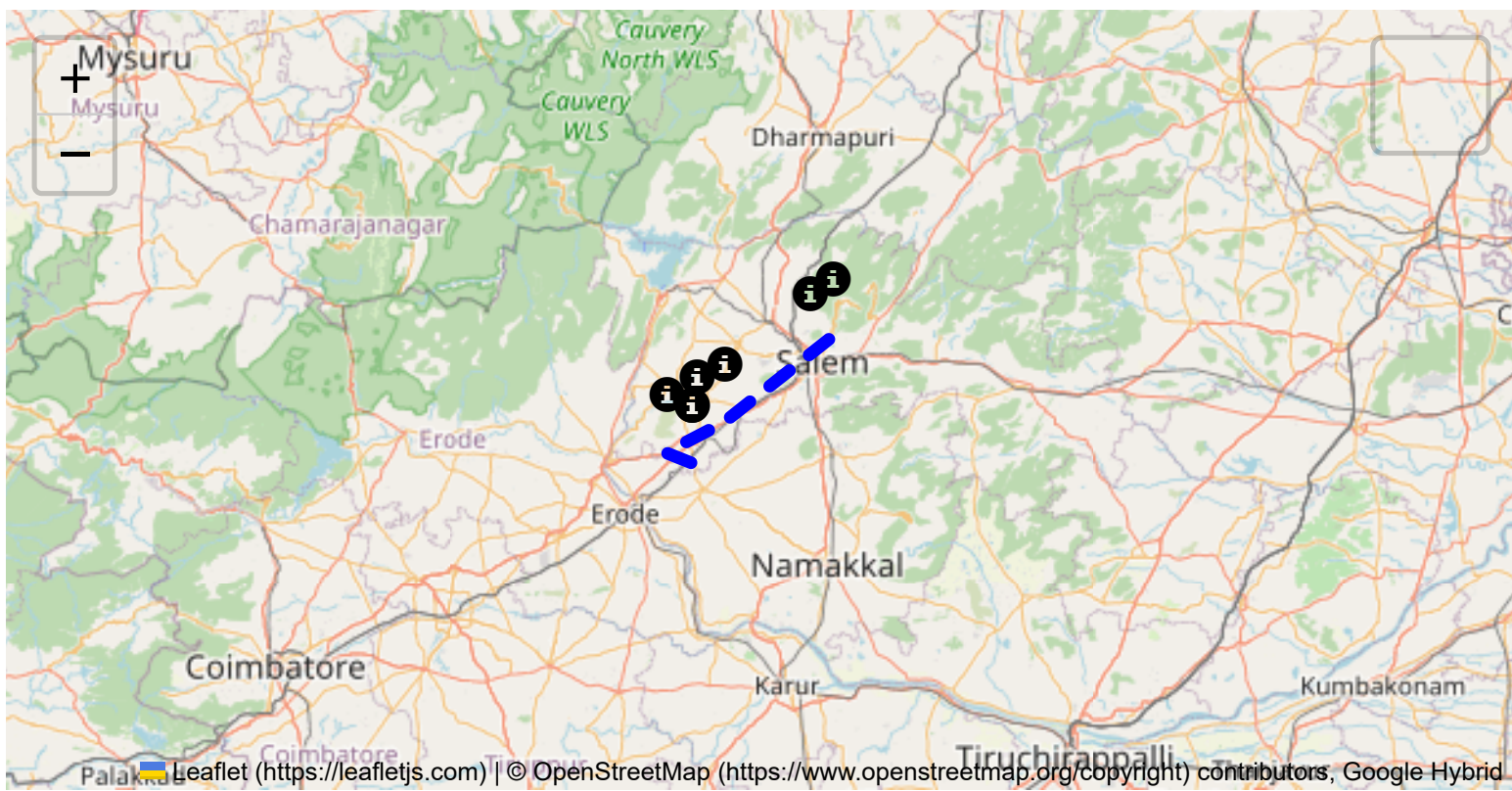
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

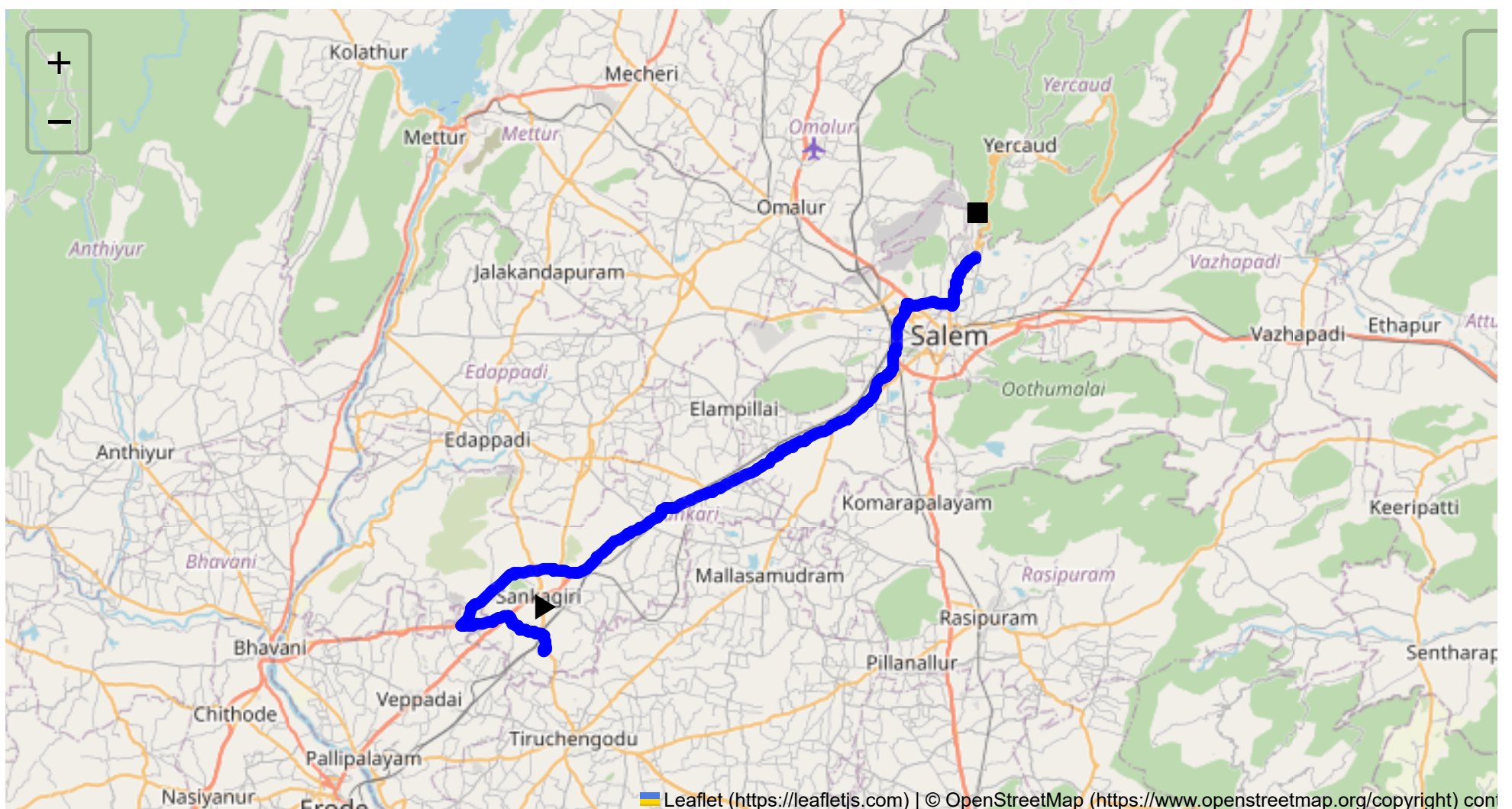
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 61.50 km
Estimated Duration: 1.3 hours
Adjusted Duration (Heavy Vehicle): 1.6 hours
Start: (11.4381, 77.8734)
End: (11.708158, 78.176083)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W in Sangagiri to 164/1A2 in Kondappanaickenpatti, Salem, covers approximately 61.50 kilometers. The journey passes through several key points including Puthur, Padaiveedu, and segments on major highways such as the Salem-Kochi Highway. The route is a mix of rural roads and busy highways, requiring careful navigation through both densely populated areas and more rural settings.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region experiences a tropical climate. Typical weather includes hot summers (March to May), a monsoon season (June to September), and mild winters (October to February). During the monsoon, the route is susceptible to heavy rains which can cause flooding and reduced visibility, increasing the risk of accidents. The summer heat can be intense, possibly leading to road surface damage like asphalt bleeding.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typically, peak hours are in the morning (7-9 AM) and evening (5-7 PM), when commuters travel to and from work.
- **Congestion-Prone Areas:** Key areas with consistent congestion include urban sections entering Salem, particularly around key city entry points and near major intersections such as the ones in Nedunchalai Nagar.

4. Assessment of Road Quality and Infrastructure

The highway sections like the Salem-Kochi Highway are generally well-maintained with proper signage and lighting, but rural roads might be narrow, with potholes and less frequent maintenance. Road quality may vary, impacting driving speed and safety for larger vehicles.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** In case of road blockages, an alternative could be taking the NH544 and NH544H, which run parallel to some of the sections of the planned route.
- **Bypass Route:** Utilizing the Salem Bypass can help avoid city congestion during peak times.

6. Summary of Local Regulations Affecting Hazardous Material Transport

In Tamil Nadu, transport of hazardous materials requires strict adherence to regulations such as proper labeling, restricted timings for transportation, and driver licensing requirements. Heavy fines apply for non-compliance. Only certain routes are approved for hazardous materials to ensure public safety.

7. Overview of Historical Incidents

Historically, the region has seen incidents involving tanker lorries due to high speed or improper handling during monsoon seasons, most prominently near the highway interchanges where heavy traffic frequency is observed. Such incidents often lead to temporary road closures and traffic diversions.

8. Environmental Considerations and Sensitive Areas

Several environmentally sensitive zones exist, including agricultural lands and small water bodies, which could be severely impacted by spills or accidents involving hazardous materials. Local fauna might be

affected as these roads also border rural and semi-forested areas.

9. Analysis of Communication Coverage

Communication coverage is generally good along the major highways, but there might be dead zones as you move into rural areas like Vaikundam. Mobile network reception can be patchy, especially during adverse weather conditions.

10. Estimated Emergency Response Times

Emergency response in urban areas like Salem can be relatively quick (15-20 minutes), but response times may increase to over 30 minutes in rural sections due to distance and road conditions. Emergency facilities are more readily available in urban centers.

11. Overall Summary of Risk Assessment

The route carries standard risks associated with a mix of urban and rural driving, with specific concerns regarding weather impact, road conditions, and traffic congestion during peak hours. Proper planning, adherence to regulations, and real-time monitoring of road conditions can mitigate many potential hazards. Drivers should be particularly vigilant about road quality fluctuations and communication issues in more remote areas.

Overall, the most significant risks involve weather-related impacts, adherence to hazardous material transport regulations, and managing urban congestion effectively. Planning for alternate routes and understanding emergency procedures can help minimize risks during the drive.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.67557, 78.15976	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.45847, 77.85140	30 KM/Hr
8	Turn	Medium	11.46048, 77.85036	30 KM/Hr
9	Turn	Medium	11.46303, 77.84945	30 KM/Hr
10	Turn	Medium	11.46314, 77.84928	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45631, 77.81668	15 KM/Hr
13	Turn	High	11.49409, 77.88004	15 KM/Hr
14	Turn	Medium	11.49419, 77.88015	30 KM/Hr
15	Turn	Medium	11.49365, 77.88596	30 KM/Hr
16	Turn	High	11.49236, 77.89552	15 KM/Hr
17	Turn	High	11.49225, 77.89560	15 KM/Hr
18	Turn	Medium	11.60942, 78.10056	30 KM/Hr
19	Turn	Medium	11.60965, 78.10070	30 KM/Hr
20	Turn	Medium	11.61736, 78.10588	30 KM/Hr
21	Turn	Medium	11.61745, 78.10601	30 KM/Hr
22	Turn	Medium	11.64310, 78.11982	30 KM/Hr
23	Turn	Medium	11.64490, 78.12024	30 KM/Hr
24	Turn	Medium	11.64503, 78.12034	30 KM/Hr
25	Turn	High	11.67074, 78.12668	15 KM/Hr
26	Turn	High	11.67081, 78.12661	15 KM/Hr
27	Turn	High	11.67582, 78.12786	15 KM/Hr
28	Turn	Medium	11.69518, 78.16364	30 KM/Hr
29	Turn	Medium	11.70719, 78.17515	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
15	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
16	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
17	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
19	hospital	Venkatesh Hospital and Urology Centre	11.6759517, 78.1324029	30 km/h	Medium
20	hospital	R.K. Hospital, Salem	11.6746905, 78.1339717	30 km/h	Medium
21	hospital	Kurinji Hospital, Salem	11.6771806, 78.1351224	30 km/h	Medium
22	clinic	Ayush Care Siddha Centre	11.6731195, 78.1351967	30 km/h	Medium
23	clinic	Bismillah Piles Clinic	11.6755285, 78.1343325	30 km/h	Medium
24	hospital	Bhavani Hospital	11.6745697, 78.1362683	30 km/h	Medium
25	clinic	Arunagiri Ent Nursing Home	11.6762823, 78.1369593	30 km/h	Medium
26	clinic	Srimaya Child Health Centre	11.6750852, 78.1370646	30 km/h	Medium
27	hospital	Gl Hospital	11.6742597, 78.1379092	30 km/h	Medium
28	clinic	Ayush Multispeciality Homoeopathic Clinic	11.6770982, 78.1395947	30 km/h	Medium
29	hospital	Balaji Skin Hospital	11.676166, 78.1379488	30 km/h	Medium
30	hospital	Dr. Sasi Eye Care	11.6743698, 78.1379234	30 km/h	Medium
31	police	Alagapuram Police station	11.6769953, 78.1422615	30 km/h	Medium
32	hospital	AP medical center	11.6774074, 78.1425988	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
33	hospital	M G Diabetes Speciality and Research Centre	11.6766146, 78.1422986	30 km/h	Medium
34	hospital	Nathan Super Speciality Hospital	11.6743874, 78.1417222	30 km/h	Medium
35	hospital	Devi Hospital	11.6786391, 78.1415731	30 km/h	Medium
36	hospital	Fairlands Eye Cure	11.677959, 78.1415897	30 km/h	Medium
38	hospital	Dr. Soodamani Hospital	11.6757031, 78.14473	30 km/h	Medium
39	hospital	Kavin Eniyan Hospital	11.677643, 78.144129	30 km/h	Medium
40	hospital	Sri Vaari Hospital	11.6763726, 78.1467277	30 km/h	Medium
41	hospital	Krishna Speciality Hospital	11.676751, 78.145561	30 km/h	Medium
44	hospital	Gopi Memorial Hospital	11.673855, 78.1516078	30 km/h	Medium
45	clinic	Saraswathi Nursing Home	11.6746327, 78.1520981	30 km/h	Medium
46	hospital	J S Hospital, Dharmapuri	11.6739064, 78.1521557	30 km/h	Medium
47	hospital	Salem Gopi Hospitals	11.6745894, 78.1515482	30 km/h	Medium
48	hospital	Dr. P. Sundarrajan Neuro Hospital	11.6742791, 78.1499738	30 km/h	Medium
49	hospital	Mahadevas Ayurvedic Hospital	11.6752045, 78.1517396	30 km/h	Medium
50	clinic	Akshaya Fertility Centre	11.6739088, 78.1518657	30 km/h	Medium
52	hospital	Shanmuga Hospital and Salem Cancer Institute	11.6770178, 78.1531844	30 km/h	Medium
53	hospital	Tms Eye Hospital	11.6772791, 78.1536202	30 km/h	Medium
54	hospital	Sureka Hospital	11.676423, 78.1550511	30 km/h	Medium
56	hospital	Ramana Surgical Hospital	11.6731205, 78.15862	30 km/h	Medium
57	hospital	Salem Medical Centre	11.67607, 78.163014	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
58	hospital	KS Siddha Hospital	11.6752685, 78.1633063	30 km/h	Medium
59	clinic	Sindhu Eye Care Centre	11.6814684, 78.1625895	30 km/h	Medium
61	hospital	Kiran Diabetic Hospital	11.6957337, 78.1625792	30 km/h	Medium
62	hospital	Malar Hospital	11.7041061, 78.16998	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
18	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
37	school	Saradha Balamandir School	11.6771182, 78.1472461	30 km/h	Medium
42	school	Saradha College	11.6769525, 78.1481901	30 km/h	Medium
43	school	Sri Sarada Matriculation Higher Secondary School	11.6768969, 78.1488115	30 km/h	Medium
51	school	Holy flower matric hr.sec school	11.6748637, 78.1528099	30 km/h	Medium
55	school	CSI Hobart Edu Institute	11.673245, 78.1567934	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
60	college	Govt ITI	11.6919383, 78.1636731	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67557, 78.15976



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49409, 77.88004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49365, 77.88596



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49236, 77.89552



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49225, 77.89560



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60942, 78.10056



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60965, 78.10070



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61736, 78.10588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61745, 78.10601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64310, 78.11982



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64490, 78.12024



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64503, 78.12034



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67074, 78.12668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67081, 78.12661



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67582, 78.12786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69518, 78.16364



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70719, 78.17515
